

EdgeTwin AI — Product & Project Document

AI-Powered Predictive Maintenance Platform (Industry 4.0)

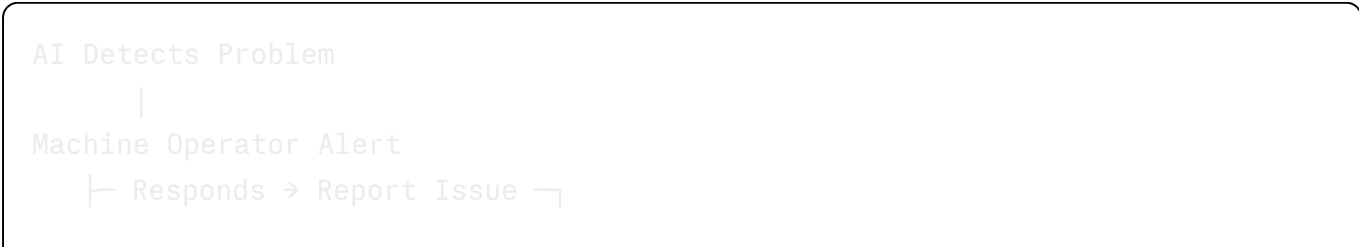
1. Overview

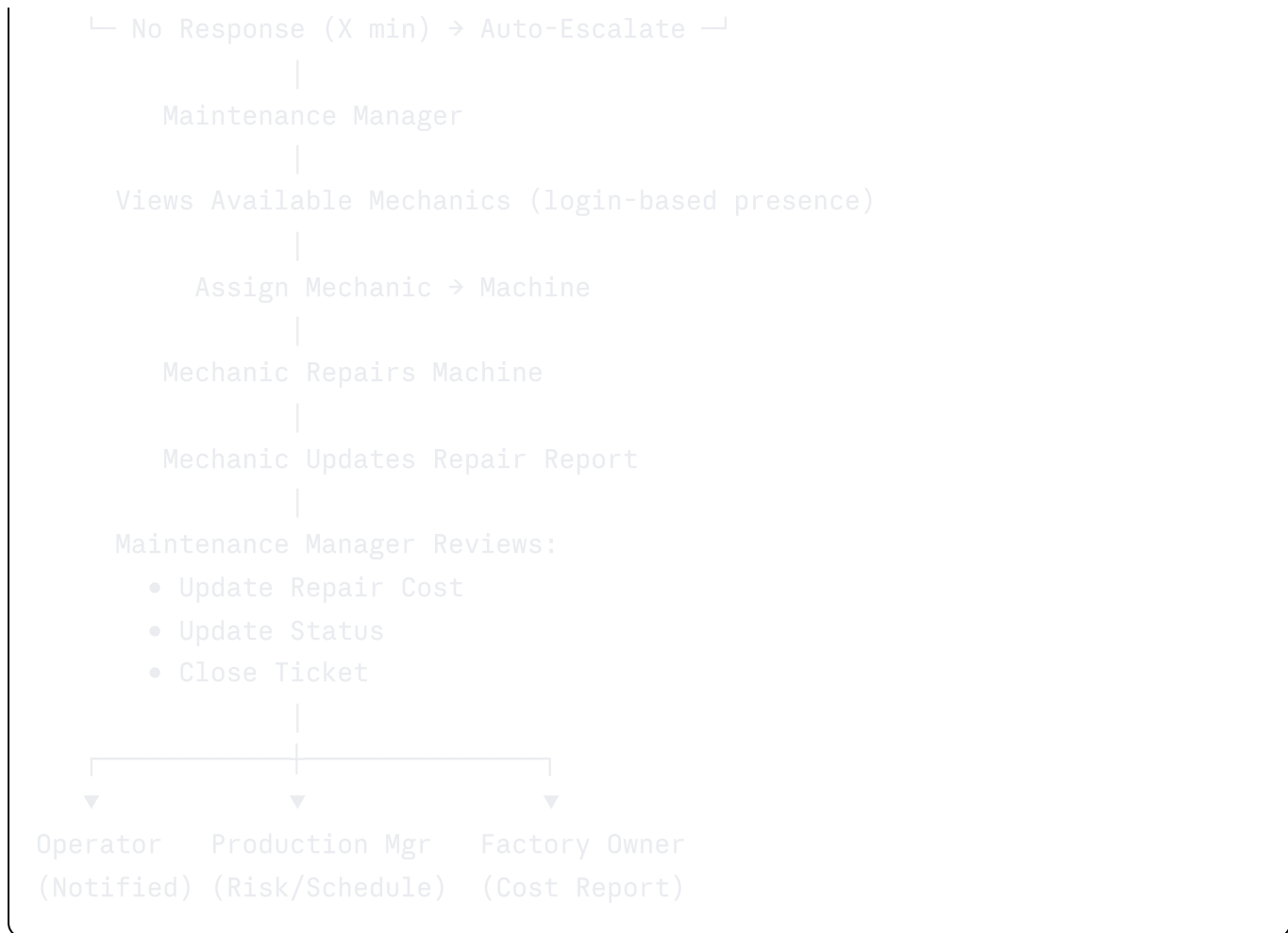
EdgeTwin AI monitors machines via sensors + ML, predicting health, risk, RUL, and failure type. It routes issues through a **multi-role escalation workflow**: Operator → Maintenance Manager → Mechanic → back to Operator/Production Manager/Factory Owner, with Admin managing users.

2. Roles & Responsibilities

Role	Core Function	Key Actions
Machine Operator	Monitors assigned machine	Report Issue / Ignore (auto-escalates)
Maintenance Manager	Central coordinator	Views mechanic availability, assigns mechanic, reviews repair report, updates cost & status, closes ticket
Mechanic (NEW)	Repairs machine	Login-based availability status, receives assignment, updates repair report on completion
Production Manager	Plans production	Views machine risk (Low/Med/High/Critical), reallocates work across machines, adjusts schedule
Factory Owner ("Principal")	Strategic oversight	Views repair cost, monthly/yearly expense graphs, critical alerts, ROI
Admin ("College Admin" analogy)	User administration	Add/remove Operators & Mechanics, roles, system config, logs

3. Updated End-to-End Flow





Business Rules (unchanged + new):

- Operator can only **Report Issue** or **Ignore**; cannot close tickets/delete alerts/edit machine.
- Auto-escalation timeout: configurable, default 5 min.
- **Mechanic availability** shown to Maintenance Manager only (based on login/shift status).
- Only Maintenance Manager assigns mechanics and closes tickets.
- Only Maintenance Manager enters **repair cost**, forwarded to Factory Owner.
- Production Manager sees **risk levels** (not raw sensor data) to plan schedules.
- Admin manages **Operator + Mechanic** accounts only (not cost/risk data).

4. New/Updated Dashboards

4.1 Mechanic Dashboard (NEW)

- Login → Availability status (Available/Busy/Offline)
- Assigned Machine & Ticket details
- Issue description, photo/voice note (from operator)
- Start Repair → Update Progress → Submit Repair Report (parts used, time taken, notes)

- History of past repairs

4.2 Maintenance Manager Dashboard (updated)

- Incoming tickets (from operator report / auto-escalation)
- **Mechanic availability panel** (real-time)
- Assign Mechanic action
- Repair report review
- Cost entry field → sync to Factory Owner
- Status update → notifies Operator + Production Manager

4.3 Production Manager Dashboard (updated)

- Machine risk matrix (Low/Med/High/Critical) across all machines
- Failure/downtime impact per line
- Reallocate work to alternate machines
- Adjust production schedule/plan

4.4 Factory Owner Dashboard (updated)

- Monthly/Yearly expense graph (maintenance cost trend)
- Critical alerts feed (cross-machine)
- Uptime/downtime, cost savings, ROI
- Repair cost breakdown by machine/period

4.5 Admin Panel (scope narrowed)

- Add/Remove Operators & Mechanics
- Assign roles, manage credentials
- Logs & activity audit
- (No access to cost/risk business data)

5. Database Schema Additions

Mechanic

```
mechanic_id, name, skill_type, login_status (available/busy/offline),  
current_assignment_id, shift, contact
```

Ticket (extended)

ticket_id, machine_id, operator_id, mechanic_id, manager_id,
status (open/assigned/in_progress/repaired/reviewed/closed),
repair_cost, repair_report, created_at, escalated (bool), closed_at

CostLog (NEW)

cost_id, ticket_id, machine_id, amount, month, year, approved_by

6. API Additions

GET /mechanics/available
POST /ticket/assign-mechanic
POST /ticket/repair-report (mechanic submits)
PUT /ticket/cost (manager updates)
PUT /ticket/close (manager only)
GET /production/risk-overview
GET /owner/expense-summary?range=month|year
GET /admin/users?role=operator|mechanic

7. Notifications Matrix

Event	Notify
Alert triggered	Operator
Ignored/timeout	Maintenance Manager (escalation)
Mechanic assigned	Mechanic, Operator
Repair report submitted	Maintenance Manager
Ticket closed + cost logged	Operator, Production Manager, Factory Owner
Critical risk detected	Production Manager, Factory Owner

8. Access Control Summary

Action	Operator	Mechanic	Maint. Mgr	Prod. Mgr	Owner	Admin
Report Issue	☒	☐	☐	☐	☐	☐
Assign Mechanic	☐	☐	☒	☐	☐	☐
Repair Report	☐	☒	☐	☐	☐	☐
Enter Cost / Close Ticket	☐	☐	☒	☐	☐	☐
View Risk Matrix	☐	☐	☒	☒	☒	☐
View Expense Graphs	☐	☐	☐	☐	☒	☐
Manage Users	☐	☐	☐	☐	☐	☒

9. Outcome

Reduced downtime · Lower maintenance cost · Higher machine life · Improved safety · Higher productivity

10. Next Steps

1. Finalize wireframes for Mechanic dashboard.
2. Extend RBAC middleware for new role.
3. Implement cost roll-up service (Ticket → CostLog → Owner analytics).
4. Add mechanic presence via WebSocket/login heartbeat.